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$$\text{compare}(a, b) = \begin{cases} 1- & \text{if } a < b \\ 0 & \text{if } a = b \\ 1 & \text{if } a > b \end{cases}$$

We can split the state into an exhaustive set of 3 disjoint cases: a < b, a = b, and a> b. Then we can define compare() using a case analysis.

1. compare(a, b)

2. case a < b

3. result := -1

4. case a = b

5. result := 0

6. else -- a > b

7. result : = 1

7.2.4 Iterative statement

An iterative process executes the same action repeatedly, subject to a condition C. If C is a condition and S is a statement, then

while C

S

is a statement, called an iterative statement, that describes the following action:

1. Test whether C is true or false.
2. If C is true, then do S and go back to step 1; otherwise do nothing.

The iterative statement is commonly known as a loop. These two steps, testing C and executing S, are repeated until C becomes false. When C becomes false, the loop ends, and the control flows to the statement next to the iterative statement. The condition C and the statement S are called the loop condition and the loop body, respectively. Testing the loop condition and executing the loop body once is called an iteration. not C is known as the termination condition.

Iterative control flow is depicted in the flowchart of Figure 7.6. Condition C has two outgoing arrows, true and false. The true arrow points to S box. If C is true, S box is executed and control flows back to C box. The false arrow points to the box after the iterative statement (dotted box). If C is false, the loop ends and the control flows to the next box after the loop.

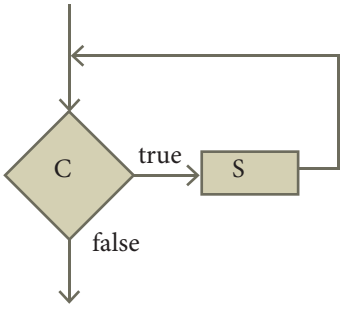

Figure 7.6: Iterative control flow

Example 7.4. Construct an iterative algorithm to compute the quotient and remainder after dividing an integer A by another integer B.

We formulated the specification of the algorithm in Example 6.6 as

divide (A , B)

-- inputs: A is an integer and B ≠ 0